CDFD Runtime Paper 02: Layered Ontology for Adaptive Systems

Steve Bico Mujjabi, MD

Independent Researcher

Founder, Vura Labs

Kampala, Uganda

ORCID: 0009-0001-0556-5516

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Abstract

Executive synthesis. The CDFD Runtime separates numerical mechanics from meaning. The engine evolves arrays; the ontology layer explains what those arrays represent in a domain. This paper upgrades the earlier "5-layer" account into a more explicit layered stack: runtime mechanics, meta-ontology, domain ontologies, application ontologies, and context-specific memory. It also explains how the discovery layer can propose new schemas, while keeping every new schema in candidate status until it survives validation.

Greek-symbol convention. Unless a manuscript states a tighter equation-local meaning, Greek symbols are local CDFD variables or coefficients, not universal constants. Here Φ denotes driving flux, Ψ_s denotes the adaptive operating ratio, Ω denotes a domain or state space, and Λ denotes the Life Number where used. The coefficients α, β, and γ denote accumulation or reinforcement, relaxation or decay, and diffusion, coupling, or redistribution. The symbols δ and ϵ denote perturbation or tolerance scales, while η, λ, μ, ν, ρ, σ, τ, and ϕ denote local efficiency, rate, baseline, frequency, density or correlation, noise or variance, time-scale, and phase or field terms. Subscripts restrict any coefficient to the named pathway, tissue, domain, or experiment.

Runtime evidence path

Prior CDFD mathematics $\longrightarrow$ CDFL/runtime state $\longrightarrow$ bounded output $\longrightarrow$ falsification target.

Figure 1: Conceptual schematic for the runtime papers. The engine translates the mathematical frame from the prior CDFD papers into executable CDFL/runtime diagnostics; candidate outputs remain tied to explicit tests before any stronger claim is made.

1 Prior CDFD Basis

This manuscript does not rederive the mathematical core of CDFD. It uses the Mujjabi Adaptive Operating Ratio and the constrained-vacuum notation introduced in Part I [1], the torus-knot hierarchy and boundary-mode work from the Part I sequence [2, 3], the public balance law developed in the Part I capstone [4], the Part I transport tests [5], and the Life Number and tri-regime biology bridge from Part II [6]. The runtime papers therefore act as an execution layer: they keep the mathematics connected to commands, outputs, falsification tests, and domain-specific limits.

2 Why Ontology Is Needed

The core engine does not know what a nephron, a market, a mineral pore, or a photonic channel is. It only evolves state fields such as Φ, C, S, M_s, and Ψ_s. That ignorance is deliberate. A reusable runtime must not hard-code every domain into the kernel.

The ontology layer supplies meaning without contaminating the engine. It maps domain concepts to runtime variables, records relationships, and exposes those relationships to CDFL, the CLI, the optional Runtime Studio, and the editor tooling that makes CDFL inspectable before execution.

3 The Runtime Ontology Stack

3.1 Layer 0: Runtime Mechanics

The runtime layer is responsible for graph mechanics, state propagation, uncertainty flags, causal links, memory binding, regime tagging, and finite diagnostic summaries. It does not decide what a domain-specific state means.

3.2 Layer 1: Core Meta-Ontology

The meta-ontology defines reusable categories:

- **Entity**: a node or bounded system.
- **Process**: a change pathway.
- **Flow**: a domain expression of Φ.
- **Constraint**: a domain expression of C.
- **Memory**: a domain expression of M_s.
- **Regime**: a qualitative label derived from runtime state.

3.3 Layer 2: Domain Ontologies

Domain adapters in `cdfd_runtime/domains` translate domain records into engine variables. The CLI currently reports 196 registered domain keys. The number matters less than the contract: each adapter must define how a domain maps to Φ and C, and must explain its interpretation limits.

3.4 Layer 3: Application Ontologies

Applications such as CDFD Studio, future scientific dashboards, or domain tools call the shared runtime backend rather than invent their own physics. This preserves one source of truth for finite checks, result envelopes, and CDFL execution.

3.5 Layer 4: Context and Memory

Context ontologies store user, project, dataset, or experiment-specific relationships. These guide interpretation, but they do not silently alter the physics kernel.

4 Schema Invention and Discovery

The local report `experiments/reports/NEW_DISCOVERIES_REPORT.md` describes a schema-invention demonstration in which a high-Ψ_s anomaly was assigned a new candidate label. The proper interpretation is not that the runtime has proven a new law of nature. The useful result is narrower: the discovery layer can detect an out-of-distribution regime, name it, record its features, and create a target for future falsification.

5 CDFL and Ontology Binding

CDFL binds ontology to execution. A `SET domain: physics` directive selects a semantic context; a `SYSTEM` block maps flow and constraint; a `RULE` block attaches an action to a threshold. The CLI now exposes this pipeline through:

```
python cdfd.py validate examples/heat_flow.cdfl
python cdfd.py run examples/heat_flow.cdfl
python cdfd.py cdfl lint examples/heat_flow.cdfl
python cdfd.py cdfl ast examples/heat_flow.cdfl --json
```

The first two commands preserve the legacy root aliases; the `cdfl` subcommands expose the language workbench explicitly.

6 Falsification Conditions

An ontology mapping fails if:

- it cannot define measurable proxies for Φ and C;
- its predicted regime thresholds do not reproduce under rerun;
- it adds no explanatory or predictive value beyond existing variables;
- it hides uncertainty behind authoritative labels.

7 Conclusion

The ontology is not decoration. It is the meaning infrastructure that lets one runtime serve many domains without pretending those domains are materially identical. Its strongest use is disciplined translation: map, run, report, falsify, and only then promote a candidate schema.

References

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